

$$\exists. Zg.: \quad \phi_n(B_n(\phi_{n-1}(B_{n-1}(\dots) + b_{n-1}) + b_n)) = A_n x + a_n$$

$\forall i \in [1, n]: \phi_i = id \downarrow$

Proof by induction:

$$n=1 : \quad \phi_1(B_1 x + b_1) \stackrel{!}{=} A_1 x + a_1$$

$$\stackrel{\phi_1 = id}{\Rightarrow} B_1 x + b_1 = A_1 x + a_1 \quad \checkmark$$

IA:

$$\forall i \in [1, n]: \phi_i = id \downarrow$$

$$\phi_n(B_n(\phi_{n-1}(B_{n-1}(\dots) + b_{n-1}) + b_n)) = A_n x + a_n$$

$n \rightarrow n+1$

$$\phi_{n+1}(B_{n+1}(\phi_n(B_n(\phi_{n-1}(B_{n-1}(\dots) + b_{n-1}) + b_n) + b_{n+1}))$$

$$\stackrel{\phi_{n+1} = id}{\Rightarrow} \phi_{n+1}(B_{n+1}(A_n x + a_n) + b_{n+1}) \stackrel{\phi_{n+1} = id}{=} \underbrace{B_{n+1} A_n x}_{= A_{n+1}} + \underbrace{B_{n+1} a_n + b_{n+1}}_{= a_{n+1}}$$

$$= A_{n+1} x + a_{n+1} \quad \checkmark$$

Comment: In comparison to the sample solution this proof uses induction rather than recursion and therefore doesn't need to absorb the bias.